MODULE – 1

1) Explain the evolution of Big Data and give a clear definition of Big Data.
2) Explain the characteristics of Big Data (Volume, Velocity, Variety, Veracity, Variability, Volatility).
3) Compare Traditional Business Intelligence (BI) and Big Data Analytics with a neat diagram.
4) What is Big Data Analytics? Explain the classification of analytics
5) Explain NoSQL databases – definition, features, types, advantages, and use cases.
6) Explain Hadoop architecture, features, advantages, and ecosystem components.

MODULE – 2

1) Compare RDBMS and Hadoop **with respect to data type, processing, scalability, cost, and use cases.**
2) Explain Hadoop architecture **with a neat diagram, highlighting** NameNode and DataNode **roles.**
3) Explain HDFS (Hadoop Distributed File System) – **features, advantages, and working.**
4) Explain MapReduce programming model **with a neat diagram and working steps.**
5) Describe YARN architecture **and explain the roles of** ResourceManager, NodeManager, ApplicationMaster, and Container**.**

MODULE – 3

1) Explain the data model of MongoDB – Database, Collection, Document, and _id with neat examples.
2) Compare RDBMS and MongoDB with respect to schema, scalability, queries, transactions, and indexing
3) Explain CRUD operations in MongoDB with suitable commands and examples.
4) Explain MongoDB data types with examples (ObjectId, Array, Embedded Document, Date, Binary, Decimal128).
5) Explain aggregation framework and MapReduce in MongoDB with suitable example queries.

MODULE – 4

1) Explain Hive architecture with a neat diagram and describe the role of Driver, Metastore, Compiler, and Execution Engine.
2) Explain Hive file formats – Text File, Sequence File, and RCFile. Compare row-oriented and column-oriented storage.
3) Explain static and dynamic partitioning in Hive with suitable examples and commands.
4) Compare Pig and Hive with respect to language, execution, performance, and use cases.

MODULE – 5

1) Explain Apache Spark architecture with a neat diagram. Describe the role of Spark Core, APIs, Data Store, and Resource Manager.
2) Explain Spark software stack with a neat diagram. Discuss the role of Spark SQL, Spark Streaming, Spark MLlib, Spark GraphX, and Spark Arrow.
3) What is Spark SQL? Explain its components, features, and advantages.
4) What are Vectorized UDFs (VUDFs) and Grouped Vectorized UDFs (GVUDFs)? Explain their working and advantages.